Solutions
QA : Algebra Review Test - 02

CEX-Q-0232/25

1	2	**2**	2	**3**	3	**4**	1	**5**	4	**6**	2	**7**	4	**8**	1	**9**	1	**10**	1
11	4	**12**	–	**13**	–	**14**	4	**15**	4	**16**	3	**17**	4	**18**	2	**19**	2	**20**	2
21	4	**22**	2	**23**	2	**24**	3	**25**	79	**26**	1	**27**	4	**28**	4	**29**	1	**30**	–

1. 2 Let us check its position for x = 0
But x = 0 makes the LHS indeterminate, which discards option (3) & (4).
Now, if we put x = 1, the inequality holds which means it is true for the values also,

So, answer: $-\frac{1}{2} \le x < \frac{45}{8}$, except x = 0.

So, option (1) also not possible.

2. 2 Put natural numbers starting with 1.
We have:
x + yz = 2
y + zx = 2
z + xy = 2
Subtracting first from second, we have,
(y – x) – z (y – x) = 0
(y – x) (1 – z) = 0
Which gives either y = x and/or z = 1.
Putting z = 1 in third equation, xy = 1 gives x = y = 1.
Hence, one solution is (1,1,1).
Further putting x = y in first/second and third equation,
x + xz = 2
and $z + x^2 = 2$
Putting $z = (2 - x^2)$ from second in first equation above, we get
$x + x(2 - x^2) = 2$
$x^3 - 3x + 2 = 0$
$(x - 1)^2(x + 2) = 0$
So, x = 1 or x = – 2.

Which gives the second only possible solution set as (– 2, – 2, – 2) which is not possible.

3. 3 The given equation can be re-arranged to give
(a + 2)(a + b + 2) = 4
The number of integral solutions of the above equation in factorized form will be the number of ways in which 4 can be expressed as product of two integers which will be 6. There are three solutions for positive products and another three solutions for negative products.

4. 1 Sum of modulus is least at the middlemost values of zero of modulus when sorted. If there are two middle values, any value from one of them to another will yield minimum.
There are 10 modulus terms in f(x).
Middlemost modulus will be from |5x – 1|, |6x – 1|

5. 4 All the terms starting from x^0 to x^{50} can be formed except for the following terms x^1, x^2, x^4, x^7, x^{47} and x^{49}. So, out of 51 powers (0 to 50), 6 are missing, hence a total of 45 terms will be formed.

6. 2 $f(x_1) = f(x_2)$ tells us that the heights of symmetric parabola are same at the two points x_1 and x_2. Because of symmetry, these two points x_1 and x_2 will be equidistant from the axis of symmetry given by x = (–b/2a) and so will be the roots. Hence,
– 5 + other root = 14 + 56
∴ Thus other root = 75
Thus, f(x) = a(x + 5)(x – 75). f(0) = a(5)(–75) while f(10) = a(15)(–65).

7. 4 No such term will exist in the expansion. Hence the coefficient will be 0, as 8 + 12 = 20 ≠ 21.

8. 1 Upto 9^2, value of each [.] will be 0. From 10^2 to 14^2 each will give 1. 15^2 to 17^2 will give 2 each. 18^2 and 19^2 will give 3 each. 20^2 to 22^2 will give 4 each. 23^2 and 24^2 will give 5 each while 25^2 will give 6 totaling (9 × 0 + 5 × 1 + 3 × 2 + 2 × 3 + 3 × 4 + 2 × 5 + 1 × 6) = 45.
So, k = 25^2 = 625 whose digital sum is 4.

9. 1 $\log_{0.3}[\log_x(2x^2 - 15x - 100)] > \log_{0.3}(2)$
$\log_x(2x^2 - 15x - 100) < (2)$
$2x^2 - 15x - 100 < x^2$
$x^2 - 15x - 100 < 0$
$(x + 5)(x - 20) < 0$
$x \in (-5, 20)$
Only positive integers greater than 1 will satisfy the given log. So, all values from 2 to 19.

10. 1 $A = (2 + 1)(2^2 + 1)(2^4 + 1) \ldots (2^{2048} + 1)$
$(2 - 1)A = (2 - 1)(2 + 1)(2^2 + 1) \ldots (2^{2048} + 1)$
$A = (2^2 - 1)(2^2 + 1) \ldots (2^{2048} + 1)$
$\Rightarrow A = (2^{2048} - 1)(2^{2048} + 1)$.
$A = 2^{4096} - 1$

$$(A+1)^{\frac{1}{2048}} = (2^{4096} - 1 + 1)^{\frac{1}{2048}} = 2^2 = 4.$$

11. 4 $\frac{P}{Q} = \frac{6a + 36d}{6a + 30d} = \frac{32}{27}$

2d = 5a.

$$S_{12} = 354 = \frac{12}{2}[a + a + 11d]$$

$$354 = 6\left[\frac{2 \times 2d}{5} + 11d\right]$$

d = 5.

12. $a^2 + 2b = b^2 + 2a + 5$
$a^2 - 2a = b^2 - 2b + 5$
$(a - 1)^2 - (b - 1)^2 = 5$
$(a + b - 2)(a - b) = 5$
$a + b - 2 = 5$
$a - b = 1$ or $a + b - 2 = 1$ or $a - b = 1$ not possible as a, b are positive.
2a = 8
a = 4, b = 3.

13. We observed that
term no from 2^1 to $2^2 - 1 = 3$ value is $2 = 2^1$
term no from 2^2 to $2^3 - 1 = 7$ value is $4 = 2^2$
term no from 2^3 to $2^4 - 1 = 15$ value is $8 = 2^3$
⋮
term no from 2^{10} to $2^{11} - 1 = 2047$ value is $2^{10} = 1024$
so at 2016th term value is = 1024 = 2^n
n = 10

14. 4 If the prose read carefully, the statement "*After attaining half the measure of his father's life chill fate took him*." it can be deduced that sum of their ages will be a multiple of 3. Only option (d) satisfies it.

Diophantus's youth lasted 1/6 of his life. He grew a beard after 1/12 more of his life. After 1/7 more of his life, Diophantus married. Five years later, he had a son. The son lived exactly half as long as his father and Diophantus died just four years after his son's death. All of this totals the years Diophantus lived.

Let *D* be the number of years Diophantus lived, and let S be the number of years his son lived. Then the above word problem gives the two equations

$$D = \left(\frac{1}{6} + \frac{1}{12} + \frac{1}{7}\right)D + 5 + 5 + 4 \qquad (1)$$

$$S = \frac{1}{2}D. \qquad (2)$$

Solving this simultaneously gives *S* = 42 as the age of the son and *D* = 84 as the age of Diophantus.

15. 4 Give two wheels to each of the 150 vehicles reducing the wheels to 300. Now, 20 wheels are left, if we give one more wheel to each of the 20 vehicles, there will be 20 auto rickshaws. So, there must be less than 20 "more than 2 wheel" vehicles. If we give 2 wheels to 10 more vehicles, those 10 will become cars. So, there must be more than 10 "more than 2 wheel" vehicles. Only (c) satisfies.
m + a + c = 150
2m + 3a + 4c = 320
Subtracting first equation from second, we get
a + 2c = 20
On checking we get for a = 4 and c = 8 or a = 6, c = 7 or a = 2, c = 9, it satisfies the equation.

16. 3 Take an example of f(x) where one of the roots is less than –1 and the other is more than 1 and cross check.
Alternative Method:
If a > 0, f(1)and f(–1) both are negative.
If a < 0, f(1) and f(–1) both are positive.
So, in either case f(1) x f(–1) > 0.
$(a + b + c)(a - b + c) > 0$
$(a + c)^2 - b^2 > 0$
$(a + c)^2 > b^2$
$(a + c)^2/b^2 > 1$

17. 4 Product of 100 positive integers is 1. Each one being equal to 1 is the only possibility. So, their sum will be 100.

18. 2 We will be trying AM ≥ GM but for that we need the expression containing terms which will cancel each other when taken into product. With a bit of intelligence, if we write it as $b + (a - b) + 1/\{b(a - b)\}$, we see that their product will yield 1.
Hence the AM of these three terms will be = product of them.

19. 2 Put random value of n = 2, 3, 4 and $[(n + 1)/2]^n$ will get farther from n!
Using AM ≥ GM
$(1 + 2 + 3 + \ldots + n)/n \geq (1 \times 2 \times 3 \times \ldots \times n)^{1/n}$
$[n(n + 1)/2]/n \geq (1 \times 2 \times 3 \times \ldots \times n)^{1/n}$
$[(n + 1)/2]^n \geq n!$
Since, n >1, $(n + 1)/2 \neq n$, and thus $[(n + 1)/2]^n > n!$

20. 2 Write down first few terms and observe the pattern

- If a_k is even, then a_{k+1} is odd, since it's of the form (even + odd).
- If a_k is odd and a_{k-1} is even, then a_{k+1} is also odd, since it's still of the same form.
- If a_k is odd and a_{k-1} is odd, then a_{k+1} is of the form (odd)(odd) + odd = even, so a_{k+1} is even.

So every third sequence term is even.
So, 998 will be even and hence 999 and 1000 will be odd.

21. 4 Write down the first few terms and check for the options. It is a series where in a group of any 3 consecutive terms, two terms are odd while one term is even. Every other option is a fact about this series.

22. 2 Put n = 1, in the options and sum given in the question and cross check, select and eliminate. Then put n = 2 and get the answer.
$S_n = t_1 + t_2 + \ldots + t_n$
$S_{n-1} = t_1 + t_2 + \ldots + t_{n-1}$
On subtracting, we get $S_n - S_{n-1} = t_n$,
or $t_n = (2n^2 - n) - [2(n-1)^2 - (n-1)] = 4n - 3$

23. 2 Use last digit check.
$(6666666666)^2 + (8888888888)$
$= [6(9999999999)/9]^2 + 8(9999999999)/9$
$= 4/9[10^{10} - 1]^2 + 8/9[10^{10} - 1]$
$= 4/9[10^{20} - 1]$
$= 4(99999999999999999999)/9$
$= (44444444444444444444)$

24. 3 All the scores possible are even. So, their sum will also be even.
Since the marking scheme is +4 and – 2, all scores from –100, – 98, – 96, …, 194, 196, 200. The score 198 cannot be scored. So, there are a total of 151 even numbers from –100 to 200 whose sum is 151x(–100 + 200)/2. So, sum of all scores is 7550 – 198.

25. 79 Rs. 79 Using two co–prime numbers 'a' and 'b', 'a × b – a – b' is the highest value that cannot be formed using them.

26. 1 Put random values which are equal, of same signs or of opposite signs and check.

27. 4 Any term will be in the form of $ka^xb^yc^zd^w$
Such that x + y + z + w = 7 and $w \le 3$ and $y \le 4$
There are total 90 possible solutions of the above equation.

28. 4 Try putting a, b, c from 1, 2 and 3 assuming first or second option is true. Then putting same as 1/1, 1/2 and 1/3 assuming either third or fourth is true.
(a + b – c)/c, (b + c – a)/a and (c + a – b)/b are in AP
(a + b – c)/c + 1, (b + c – a)/a + 1 and (c + a – b)/b + 1 are in AP
(a + b + c)/c, (b + c + a)/a and (c + a + b)/b are in AP
1/c, 1/a and 1/b are in AP
c, a and b are in HP

29. 1 From given data, we can say that
f(x) = k(x – 1) (x – 2) (x – 3) .....(x – 49) + x + 3
So, f(50) = k49! + 53.

30. $x + \frac{1}{y} = y + \frac{1}{z} = z + \frac{1}{x}$

solving in pair we get

$x - y = \frac{y-z}{yz}$ … (i)

$y - z = \frac{z-x}{zx}$ … (ii)

$z - x = \frac{x-y}{xy}$ … (iii)

(i) × (ii) × (iii)

$(x-y)(y-z)(z-x) = \frac{(x-y)(y-z)(z-x)}{x^2y^2z^2}$

$\therefore x^2y^2z^2 = 1.$